



WOW!

The 2011 Japanese tsunami raised the sea level by 30 ft (9 m) at Miyako in northeast Japan, and sent waves 6 miles (10 km) inland.

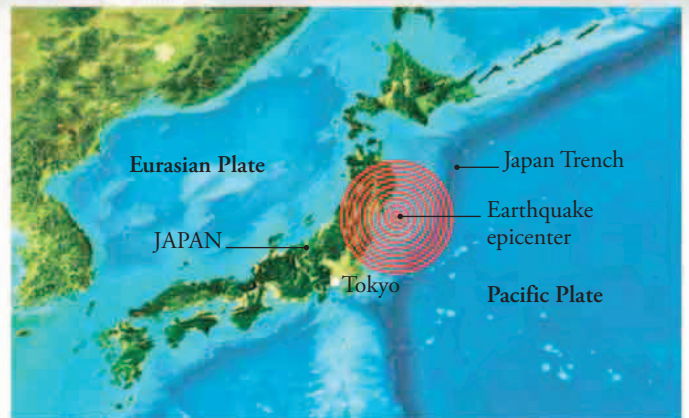


LANDFALL

When a tsunami reaches shallower water, the wave shortens and steepens. This creates a very high but broad wave peak and an equally deep trough. The trough usually reaches the coast first, making the sea draw back like a very low tide. But this is soon followed by the tsunami peak, which surges ashore and floods the landscape.

▲ TSUNAMI SURGE

The relentlessly rising water of the 2011 Japanese tsunami surges over the sea wall at Miyako.



DISASTER ZONE

As the tsunami rushes over the land, the water behaves like a giant liquid bulldozer, destroying everything in its path. The moving water becomes thick with floating debris, including sea-going ships that often end up stranded in the middle of wrecked coastal cities.

POWERFUL EARTHQUAKE

During the earthquake that caused the 2011 Japanese tsunami, the Pacific floor slipped west into the Japan Trench by more than 66 ft (20 m). Meanwhile, the main island of Japan shifted eastward by 8 ft (2.4 m). A long stretch of its eastern coast sank by 2 ft (60 cm), and this allowed the tsunami waves to flood even more of the land. Some of the debris that was swept out to sea drifted across the Pacific Ocean as far as the US.